PayTrackEdu

Business Logic Guide — how the institution manages enrolments and tuition payments

PayTrackEdu supports an educational institution from academic planning through student enrolment, scheduled fees, collection of payments, and balance follow-up. It keeps the financial history linked to each student's enrolment.

1. The business cycle

Academic year + promotion → promotion session → payment plan → student enrolment → instalments → payments → balance tracking

The institution first defines what it offers, then enrols students into a specific offering for a specific academic year. A chosen payment plan determines what the student owes and when. Every received payment is assigned to the student's outstanding instalments, making the current balance visible at all times.

2. Academic setup

Academic years

An academic year has a unique code, start date, end date, and lifecycle status: Draft, Active, Closed, or Archived. The end date cannot be earlier than the start date.

Promotions

A promotion represents an educational programme, class, or cohort definition. Its code and name must be unique. A promotion can be active or inactive.

Promotion sessions

A promotion session connects one promotion to one academic year. For example, a "Web Development" promotion can have a separate session for the 2026-2027 academic year. The system automatically gives the session a combined code and records its registration fee.

Business rule: the same promotion can have only one active session in a given academic year.

A session moves through Draft, Open, Closed, and Archived. The status communicates whether it is being prepared, open for enrolment, closed, or retained for history.

3. Students and enrolments

Student records

Each student record stores identity, contact information, date of birth, and parent contact information. An active student email must be unique. Records are archived rather than permanently removed, so history can be retained and restored when needed.

Creating an enrolment

An enrolment links exactly three things: a student, a promotion session, and a payment plan belonging to that same session. It begins in **Draft** status.

1. Select an active student.

2. Select the academic-year promotion session.
3. Select one of that session's payment plans.
4. Optionally enter a discount and its reason.
5. The system stores the session's registration fee and the plan's tuition amount on the enrolment.
6. The system automatically creates the enrolment instalments.

Business rules: a student cannot have two active enrolments in the same promotion session; a plan cannot be selected for a different session; and a discount cannot exceed the full registration-plus-tuition total.

Enrolment status

Status	Meaning
Draft	The enrolment has been prepared but is not yet active.
Active	The student is currently enrolled and payments can be recorded.
Suspended	The enrolment is temporarily paused.
Cancelled	The enrolment is no longer valid; no new payment can be created for it.
Completed	The enrolment is finished. It cannot later be activated, suspended, or cancelled.

4. Payment plans and instalments

A payment plan defines how the tuition for a promotion session is paid. The supported plan types are Cash, Trimester, and Monthly. Each plan has a total tuition amount and one or more dated plan instalments.

A plan instalment is either a **deposit** or a **tuition** instalment. Its sequence number must be unique within its plan, so the payment order is clear.

How an enrolment schedule is created

When an enrolment is created, PayTrackEdu creates a personal schedule for that student:

- First, a registration-fee instalment is created with today's date.
- Then every instalment in the selected payment plan is copied into the enrolment schedule, with its amount and due date.
- Tuition entries are labelled in order, such as "Tuition 1" and "Tuition 2".

Discount treatment: a discount is applied from the last tuition instalment backwards. If it is larger than tuition, any remaining discount then reduces the registration fee. No instalment is reduced below zero.

Staff may change the due date of a non-cancelled instalment or cancel an instalment. A cancelled instalment is excluded from the amount that remains payable.

5. Payments and automatic allocation

A payment is recorded against an enrolment and includes its amount, payment date, method (Cash, Check, or Bank Transfer), optional reference, optional notes, and an automatically generated receipt number.

Business rules: a payment must be greater than zero, cannot be recorded for a cancelled enrolment, and cannot exceed the remaining amount due for that enrolment.

Automatic allocation

Immediately after a payment is recorded, the system allocates it to outstanding non-cancelled instalments in their schedule order. It pays the earliest unpaid instalment first, then continues with the next one until the payment is fully used or no balance remains.

Example: if the registration fee has 100 remaining and Tuition 1 has 250 remaining, a payment of 220 is allocated as 100 to registration and 120 to Tuition 1. The payment is fully accounted for, and Tuition 1 has 130 still due.

Manual allocation

Staff can also allocate part of a payment to a particular instalment. The payment and instalment must belong to the same enrolment. The allocated amount cannot exceed either the unallocated part of the payment or the outstanding balance of that instalment.

Instalment status

Status	When it applies
Pending	No active payment has been allocated and the due date has not passed.
Partially paid	Some, but not all, of the expected amount has been paid.
Paid	Active allocations meet or exceed the expected amount.
Overdue	There is still an unpaid amount after the due date.
Cancelled	The instalment has been cancelled and is no longer payable.

6. Balances, cancellation, and auditability

Balance calculations

For each enrolment, the system shows the following values:

- **Total amount** = registration fee + tuition amount.
- **Final amount** = total amount - discount.
- **Paid amount** = allocations from active payments only.
- **Remaining amount** = final amount - paid amount.

Cancelled instalments do not form part of the remaining payable schedule. This prevents the institution from collecting payment for a charge it has cancelled.

Cancelling a payment

A payment is not deleted. It is marked as cancelled and records who cancelled it, when, and why. Its allocations stop contributing to paid totals. The linked instalments are then recalculated, so a previously paid instalment can become partially paid, pending, or overdue again.

> **Business rule:** a cancelled payment cannot be allocated again.

Record retention

Academic years, promotions, promotion sessions, students, payment plans, and plan instalments are archived through soft deletion rather than erased. Archived records disappear from normal active lists but can be restored. Payments preserve their financial history through status changes rather than deletion.

7. Users and everyday operation

The system identifies Admin and Staff users. A signed-in user is required to operate the protected business functions. A typical daily workflow is:

1. Maintain academic years, promotions, sessions, and payment plans.
2. Create or update student records.
3. Enrol each student into a session with the appropriate payment plan and any approved discount.
4. Review the generated schedule and adjust due dates only when justified.
5. Record each received payment and use the generated receipt as its reference.
6. Monitor unpaid, partially paid, and overdue instalments; cancel payments or charges only with a recorded reason where required.

Prepared from the current PayTrackEdu application rules. This guide describes business behaviour, not technical implementation.